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THERAPEUTIC

News Letter

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MOU Between Chalapathi Institute of Pharmaceutical Sciences and Govt. General Hospital, Guntur (30-12-2008)

CIPS-Chalapathi Institute of Pharmaceutical sciences was started in the year 2004. The institute has gained respect of the peers in academic and industry for the quality of academic and research activities being carried out.

In order to interconnect physicians, other health care team and budding pharmacists, a triannual magazine "Department of Pharmacy Practice, Chalapathi Institute of Pharmaceutical Sciences- *Therapeutic newsletter*" is being started both in electronic and print version.

Therapeutic newsletter is designed to meet the changing needs of patient care. It is intended for use not only by pharmacists but also by the persons, who are already practicing in patient care settings, and by new students – the pharmacists of tomorrow.

Pharmacists have mostly been involved in the industry aspect rather than health care. With the introduction of Pharm.D the knowledge base of pharmacy graduates is changing, to provide better patient care by co-operative working relationship between pharmacist and other health care professionals ultimately to improve the patients quality of life.

The pharmacist's task is to ensure that a patient's drug therapy is appropriately indicated, the most effective available, the safest possible, and convenient for the patient.

"Pharmaceutical care is the direct, responsible provision of medication-related care for the purpose of achieving definite outcomes that improve a patient's quality of life"

Further Information, Department of Pharmacy Practice



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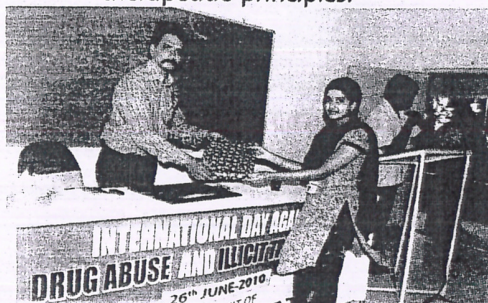
KNOW ABOUT PHARMACY PRACTICE

Pharmacy practice is the discipline of pharmacy which involves developing the professional roles of pharmacists. It is concerned with the promotion of safe, effective and economic

use of medication. Pharmacist primarily supervises medication distribution and counsels the patient by providing patient centered medication therapy management, disease-state management and clinical intervention.

Objectives

- ▼ To provide the information about Drug and New therapeutic agents.
- ▼ To provide Patient Counselling.
- ▼ To monitor and assess Adverse Drug Reactions.
- ▼ To perform Therapeutic Drug Monitoring.
- ▼ To provide patient care in co-operation with physician and other health care team.
- ▼ To ensure the Rational Drug Therapy.
- ▼ To provide Evidence Based Medication by using updated therapeutic principles.
- ▼ To monitor and assess Drug Interactions.
- ▼ To provide safe and effective drug therapy to special population.
- ▼ To avoid unnecessary polypharmacy.
- ▼ To dispense prescription drug orders.
- ▼ To provide the Poison Information Services.
- ▼ To conduct CME (Continuing Medical Education) to paramedical and medical students.
- ▼ To carryout pharmaco-economic study.
- ▼ Research and Development.



Functions

DEPARTMENT OF PHARMACY PRACTICE

- ▼ Patient counselling: providing information and assistance to the patients regarding the appropriate use of medications
 - ▶ Providing labeling of medications with directions for use
 - ▶ Advising patients about possible drug interactions
 - ▶ Educating patients in their understandable language and by providing patient leaflets
- ▼ Drug Information Services
 - ▶ Providing Information regarding drugs directly or over phone and responding to queries through website
 - ▶ Providing Poison Information to Emergency Department
 - ▶ Following up and documentation of queries
- ▼ Prescription handling and dispensing of medication
- ▼ Review of prescription for ADR and Drug Interactions
- ▼ Preparation of Drug Formulary
- ▼ Review of patient medical and medication history
- ▼ Regular visits to wards for monitoring completeness and accuracy in adherence to the drug therapy
- ▼ Pharmacist is available for consultation by medical and nursing staff and ensures that the drug distribution is operating correctly
- ▼ Interpretation of Therapeutic Drug Monitoring and ensuring safe use of the drugs
- ▼ Monitoring and reporting of adverse events to Pharmacovigilance Department
- ▼ Preparation of Patient Information leaflet
- ▼ Participating in Infection Control and Monitoring Committee
- ▼ Proper storage and dispensing of medicines

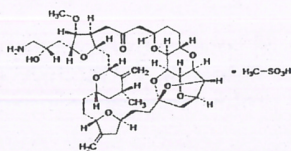


ERIBULIN MESYLATE (HALAVEN)

Drug profile

Eribulin Mesylate is a non-taxane microtubule dynamics inhibitor.

Eribulin mesylate is a synthetic analogue of halichondrin B, a product isolated from the marine sponge *Halichondria okadae*. The chemical name for Eribulin Mesylate is 11,15:18,21:24,28-Triepoxy-7,9-ethano-12,15-methano-9*H*, 15*H*-furo[3,2-*f*] furo [2',3':5,6] pyrano [4,3-*b*] [1,4] dioxacyclopentacosin-5(4*H*)-one, 2-[(2*S*)-3-amino-2-hydroxypropyl] hexacosahydro-3-methoxy-26-methyl-20,27-bis (methylene)-, (2*R*,3*R*,3*aS*,7*R*,8*aS*,9*S*, 10*aR*, 11*S*, 12*R*, 13*aR*,13*bS*, 15*S*,18*S*, 21*S*,24*S*,26*R*, 28*R*,29*aS*)-methanesulfonate (salt). It has a molecular weight of 826.0 (729.9 for free base). The empirical formula is $C_{40}H_{59}NO_{11} \cdot CH_4O_3S$. Eribulin mesylate has the following structural formula:



CLINICAL PHARMACOLOGY

Mechanism of Action

Eribulin inhibits the growth phase of microtubules without affecting the shortening phase and sequesters tubulin into nonproductive aggregates. Eribulin exerts its effects via a tubulin-based antimetabolic mechanism leading to G₂/M cell-cycle block, disruption of mitotic spindles, and, ultimately, apoptotic cell death after prolonged mitotic blockage.

Pharmacokinetics

The pharmacokinetics of eribulin is linear with a mean elimination half-life of approximately 40 hours, a mean volume of distribution of 43 L/m² to 114 L/m² and mean clearance of 1.16 L/hr/m² to 2.42 L/hr/m² over the dose range of 0.25 mg/m² to 4.0 mg/m². The human plasma protein binding of eribulin at concentrations of 100 ng/mL to 1,000 ng/mL ranges from 49% to 65%. Eribulin exposure after multiple dosing is comparable to that following a single dose. No accumulation of Eribulin is observed with weekly administration.

Metabolism

Cytochrome P450 3A4 (CYP3A4) negligibly metabolizes Eribulin *in-vitro*. Eribulin inhibits CYP3A4 activity in human liver microsomes, but it is unlikely that Eribulin will substantially increase the plasma levels of CYP3A4 substrates. Eribulin shows no induction potential for CYP1A, CYP2C9, CYP2C19, and CYP3A in primary human hepatocytes. No significant inhibition of CYP1A2, CYP2C9, CYP2C19, CYP2D6, or CYP2E1 was detected with eribulin concentrations up to 5 μM in pooled human liver microsomes. *In vitro* drug interaction studies indicate that eribulin does not inhibit drugs that are substrates of these enzymes and it is unlikely that eribulin will affect plasma levels of drugs that are substrates of CYP enzymes. Eribulin is a substrate and a weak inhibitor of the drug efflux transporter P-gp *in vitro*.

Elimination

Eribulin is eliminated primarily in feces unchanged. After administration of ¹⁴C-eribulin to patients, approximately 82% of the dose was eliminated in feces and 9% in urine. Unchanged eribulin accounted for approximately 88% and 91% of the dose in feces and urine, respectively.

INDICATIONS

Eribulin is indicated for the treatment of patients with metastatic breast cancer who have previously received at least two chemotherapeutic regimens for the treatment of metastatic disease. Prior therapy should have included an anthracycline and a taxane in either the adjuvant or metastatic setting.

DOSAGE AND ADMINISTRATION

Recommended Dose

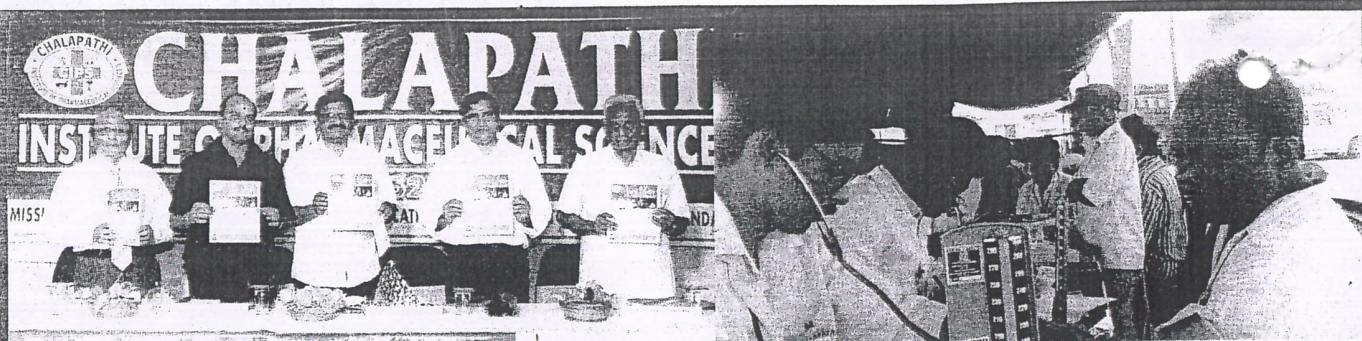
The recommended dose of HALAVEN is 1.4 mg/m² administered intravenously over 2 to 5 minutes on Days 1 and 8 of a 21-day cycle.

Patients with hepatic impairment is 0.7 mg/m² administered intravenously over 2 to 5 minutes on Days 1 and 8 of a 21-day cycle.

Patients with renal impairment (creatinine clearance of 30-50 mL/min) is 1.1 mg/m² administered intravenously over 2 to 5 minutes on Days 1 and 8 of a 21-day cycle.

Instructions for Preparation and Administration

Aseptically withdraw the required amount of HALAVEN from the single-use vial and administer undiluted or diluted in 100 mL of 0.9% Sodium Chloride Injection, USP.



Do not dilute in or administer through an intravenous line containing solutions with dextrose.

Do not administer in the same intravenous line concurrent with the other medicinal products.

Discard unused portions of the vial

SIDE EFFECTS

The common adverse reactions were neutropenia, anemia, asthenia/fatigue, alopecia, peripheral neuropathy, nausea, and constipation.

DRUG INTERACTIONS

Eribulin does not inhibit CYP1A2, CYP2C9, CYP2C19, CYP2D6, CYP2E1 or CYP3A4 enzymes or induce CYP1A2, CYP2C9, CYP2C19 or CYP3A4 enzymes at relevant clinical concentrations. Eribulin is not expected to alter the plasma

concentrations of drugs that are substrates of these enzymes

PRECAUTIONS

Pregnancy Category D

OVERDOSE

Overdosage of Eribulin has been reported at approximately 4 times the recommended dose, which resulted in Grade 3 neutropenia lasting seven days and a Grade 3 hypersensitivity reaction lasting one day. There is no known antidote for Eribulin overdose.

HOW SUPPLIED

HALAVEN (Eribulin Mesylate) Injection, 1 mg/2 mL (0.5 mg/mL). Store at 25°C (77°F); excursions permitted to 15° - 30° C (59° - 86° F). Do not freeze. Store the vials in their original cartons.

DOCTOR OF PHARMACY (PHARM.D)

1. What is Pharm.D course?

Pharm.D course is a Professional Pharmacy doctoral programme of 6 yrs duration after 10+2 (science academic stream) which includes 5yrs of academic study + 1 year of internship or residency.

2. What are the eligibility criteria to join Pharm.D courses?

- A pass in 10+2 examination with Physics and Chemistry as compulsory subjects along with one of the following subjects: Mathematics or Biology.
- A pass in D.Pharm course from an institution approved by the Pharmacy council of India under section 12 of The Pharmacy Act.

3. In what way Pharm.D differs from regular Pharmacy course?

Pharm D programme emphasizes Pharmacy practice component in all spheres of Pharmacy professional services like Hospital & Clinical Pharmacy, Community Pharmacy, Clinical Research, Regulatory Affairs, Formulation Development, quality control and after Pharm D qualification the professionals are eligible to undertake Ph.D. programme.

4. What is the scope of Pharm.D?

They can enter in the field of Clinical Pharmacy, Hospital Pharmacy, Community Pharmacy, Clinical Research, Intellectual property Rights (IPR), Regulatory Affairs, Research & Development (R&D), Formulation and Development (F&D), Production, Marketing, Lectureship and many more in India and Abroad (USA, UK, Australia etc).

5. Can I prefix Dr to my name?

Yes, for the purpose of practicing Pharmacy Profession under Pharmacy Act.

CHALAPATHI INSTITUTE OF PHARMACEUTICAL SCIENCES

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